

Multidimensional Goodhart

Response Channels, Scorecards, and Residual Shape

Iteration 47 draft — seven-part spine

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1 The Licensing Problem

1.1 The warning and what it does not identify

Goodhart’s law is usually introduced through the compressed formulation “When a measure becomes a target, it ceases to be a good measure” [20]. The original macroeconomic warning is sharper about policy use: “Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes” [7]. Both formulations describe a real empirical phenomenon: once a measure is used for control, the residual error between proxy and target can acquire direction, support, active constraints, exchange rates, tail behavior, and response dynamics. But none of those shapes is identified by the claim that the proxy became a target.

The same observed score improvement can come from several mechanisms. A fixed pool can be re-ranked so better-scoring units enter the selected set. Continuing units can repair measurement errors with no hidden harm. The same units can game a high-score, high-harm action channel. Or the score can improve because the target itself improved. The aggregate score path does not distinguish those stories. A calculation is licensed only after the response model says what is fixed, what changes, and what hidden value or harm is being measured.

This book treats “multidimensional Goodhart” as a response-modeling framework, not as a single universal theorem. The framework asks what proxy pressure changes and which primitives are declared before importing a bound.

1.2 What “multidimensional” actually adds

Let S be a state space. A target map $G : S \rightarrow \mathbb{R}^m$ records target-relevant state and a proxy map $P : S \rightarrow \mathbb{R}^k$ records measured features. The intended correspondence is $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$, with residual $\varepsilon(s) = P(s) - \varphi(G(s))$.

Two gaps matter. The **dimension gap** is the part of target variation the proxy map cannot see, represented by directions such as $\ker \varphi$. The **observation gap** is residual measurement artifact inside the measured domain, represented by ε . A scalar score may hide both gaps at once.

Two response channels matter as much as the geometry. Here $u \in U$ is a declared fixed type: the agent, unit, hospital, model checkpoint, or institution as represented before the policy response being studied. It contains the attributes treated as fixed for the comparison; choices made because of the policy are not part of u . The baseline type law is ν . A response kernel $K_{\theta(ds | u)}$ is the conditional law of the observed state $s \in S$ for a unit of type u after policy exposure θ . A participation or selection weight $W_{\theta(u)} \geq 0$ changes how much type u appears in the realized population. With these objects,

$$\mu_{\theta(A)} = \frac{\int W_{\theta(u)} K_{\theta(A | u)} \nu(du)}{\int W_{\theta(u)} \nu(du)}.$$

Pure selection means $K_{\theta} = K_0$ for ν -almost every type and policy dependence enters only through W_{θ} . Intervention means K_{θ} changes on a positive- ν set of fixed types.

Selection and intervention can produce similar marginal score movement, so they do not license the same mathematics.

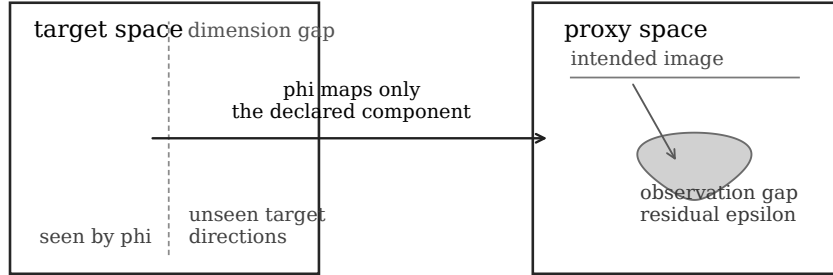


Figure 1: Dimension and observation gaps are different contract fields. The schematic licenses the split in vocabulary, not a quantitative conclusion.

1.3 What we tried and what failed

The negative results are not housekeeping; they are the reason the framework has its current shape.

Claim gallery.

- **Unconditional dimensional scaling fails.** Hidden harm does not scale with $\dim(\ker \varphi)$ by itself. If hidden coordinates are independent of the selected proxy, thresholding does not move them.
- **“More measured dimensions means more error” has no sign.** Additive and conjunctive aggregation can give opposite comparative statics from the same primitive action channels.
- **Signed aggregate hidden error is unstable.** Positive and negative hidden movements can cancel. A value functional, norm, tail risk, or domain loss has to be declared.
- **Covariance is not a finite-pressure primitive.** With $P = Z$ and $H = Z^2 - 1$, baseline covariance is zero but threshold or Boltzmann selection moves H .
- **Absolute continuity is not the intervention boundary.** It is useful for the reweighting theorem, but fixed-type action changes can remain absolutely continuous with respect to the baseline law.
- **The selection/intervention split is not marginally identifiable.** It is relative to a declared type/action representation and needs repeated-type, exposure, action-trace, or structural evidence.
- **The selection bound is not coordinate-free by itself.** It is coordinate-explicit until a value norm or scalar value functional is named.
- **Convex affordability is not welfare.** A score-deficit cost bounds private action affordability, not hidden harm.
- **Minimum-complexity attraction is not generic.** Selection follows baseline tails; intervention follows cost, search, caps, fixed charges, and affordances.

These failures block a single “n-dimensional Goodhart law.” What survives is a set of conditional calculations plus a discipline for saying when each calculation applies.

1.4 The response-modeling contract

A Goodhart claim must declare the primitives that make it a claim rather than a retrospective label:

Contract.

- **Claimed output:** the object the application wants to compute or classify: scalar score drift, hidden drift vector, induced distribution, selection/intervention label, intervention feasibility bound, welfare comparison, or response-shape prediction.
- **Type representation:** the measurable type space U , interpreted as the population description: what one element $u \in U$ represents, which attributes of that unit are fixed for the comparison, and which variables are instead actions, noise, outcomes, or later states.
- **Baseline behavior:** the baseline type law ν and baseline response kernel $K_0(ds \mid u)$: the conditional distribution of observed states before the policy response being modeled.
- **Policy exposure:** the rule, threshold, ranking, scorecard, or incentive indexed by θ , and who can observe or respond to it.
- **Response channel:** whether θ changes only selection weights $W_{\theta(u)}$ at fixed K_0 , changes the fixed-type response kernel $K_{\theta(ds \mid u)}$, or uses an explicit mixture. Pure selection is $K_\theta = K_0$ ν -almost surely; intervention is kernel change on a positive- ν set of types.
- **Action/search geometry:** available actions, costs, caps, fixed charges, search process, stakes V , and how those actions move the state distribution when an intervention bound is imported.
- **Proxy/target relation:** the maps P , G , φ , residual ε , and the relevant dimension and observation gaps.
- **Aggregation:** additive, conjunctive, threshold, Pareto, lexicographic, or institution-specific rule.
- **Hidden value or harm:** coordinates, scalar value vector, norm, or harm rates h_j .
- **Evidence standard and falsifier:** observations that would distinguish the claimed channel from nearby alternatives and observations that would make the import fail.

The first row is load-bearing. The contract is not a single formula with one fixed output. It is an adequacy test for a proposed output. A scalar hidden drift claim needs less information than a full response kernel. A distribution over k discrete states has $k - 1$ degrees of freedom. A hidden drift vector in $\mathbb{R}^m$ needs m value-relevant coordinates. A response-kernel claim can be infinite-dimensional unless the model restricts it. A selection/intervention label has a small output alphabet, but it is causal: it cannot usually be read from the marginal score path.

The type representation is part of the empirical claim, not notation to hide inside the model. If U is too rich, it can encode each unit’s whole policy-contingent response

plan and make every intervention look like selection. If U is too coarse, stable heterogeneity can look like a kernel change. The contract therefore has to defend why the chosen u is fixed for the comparison and why omitted variation belongs in K_θ , W_θ , or the action model.

The kernel language should be read conditionally. $K_{\theta(B|u)}$ is the probability that a fixed type u produces an observed state in event $B \subset S$ after exposure θ . A change in W_θ changes which fixed types are represented; a change in K_θ changes what a fixed type does or produces. The induced marginal law μ_θ usually cannot distinguish those stories by itself.

After the output is named, the contract should pass a small information accounting check:

Contract adequacy.

1. State the output object and its degrees of freedom or identifying data.
2. List the raw primitives the claim uses: type law, kernels, weights, proxy and target maps, aggregation, action costs, stakes, hidden value, and evidence.
3. Subtract constraints, such as probability normalization, declared equality $K_\theta = K_0$, additive aggregation, convex cost, or fixed deficit.
4. Subtract redundancies, such as the arbitrary scale of W_θ or coordinate rescalings that leave the represented situation unchanged.
5. Check units: costs must be comparable to stakes, proxy gains to score deficits, and hidden harm rates to the declared value units.
6. Check rank or relevance: independent changes in the declared inputs should affect the claimed output, unless the redundancy has been stated.
7. Check invariance: the conclusion should not depend on arbitrary naming, coordinate splitting, units, or post-hoc enrichment/coarsening of U .

In shorthand, a contract is adequate only if

$$\text{true input information} = \text{raw variables} - \text{constraints} - \text{redundancies}$$

is enough to identify the claimed output up to the intended invariances. If two meaningfully different response stories still satisfy the same declared inputs, the contract has not licensed that output. It may still license a weaker output: for example, an aggregate score path may license a monitoring trigger while not licensing a welfare verdict or a selection/intervention classification.

The contract is demanding by design. It prevents the framework from inferring welfare, hidden target movement, or a response channel from marginal score movement alone.

2 Licensed Calculations

2.1 Selection channel

This section covers T1 and T2 in `research/core-math.md`.

Before the theorem can be used, the contract must supply a baseline law, a pure selection channel, hidden coordinates or a value metric, and enough integrable moments. Pure selection means $K_\theta = K_0$ and the policy changes only the weights. In the induced marginal law, write $L = d^{\frac{\mu_\theta}{d}} \mu_0$ and $\delta = \|L - 1\|_{L^2(\mu_0)}$.

T1. Coordinate-explicit selection bound. For hidden coordinates H_i with finite second moments and baseline standard deviations s_i ,

$$|\mathbb{E}_{\mu_\theta}[H_i] - \mathbb{E}_{\mu_0}[H_i]| \leq \delta s_i$$

for each coordinate. After a Euclidean coordinate metric is declared,

$$\|B_{H(\theta)}\|_2 \leq \delta \|s\|_2$$

.

T2. Value-weighted/operator selection bound. For a declared scalar value direction v and covariance Σ_H ,

$$|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}$$

. For a declared norm,

$$\|B_{H(\theta)}\|_V \leq \delta \sup_{\|v\|_{V,*} \leq 1} \sqrt{v^T \Sigma_H v}$$

.

These are Hilbert-space Cauchy–Schwarz bounds. They license drift envelopes for pure reweighting. They do not identify the hidden coordinates, the value weights, or the welfare object. They do not apply to fixed-type response changes.

Covariance belongs here only as a local velocity. Along a valid exponential tilt, the derivative at zero pressure is a covariance. At finite pressure, the path response, tail shape, and moment-generating domain matter. That is why the zero-covariance example $H = Z^2 - 1$ survives as a warning.

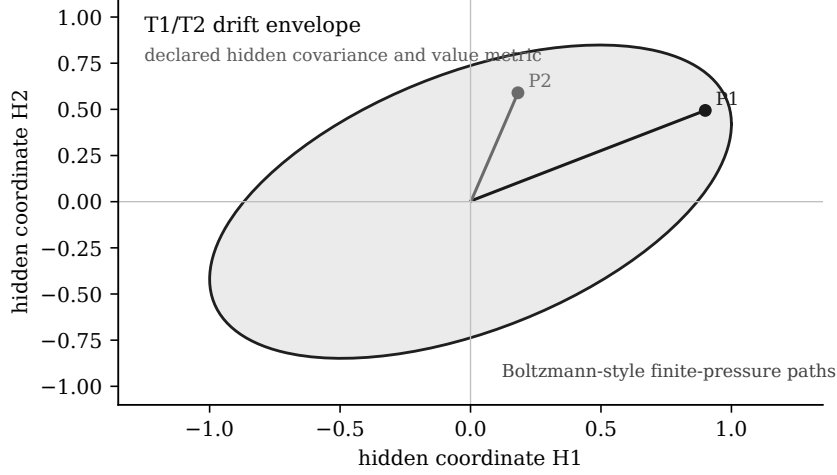


Figure 2: T1/T2 give a declared hidden-space envelope. Boltzmann-style pressure paths are trajectories inside that declared geometry, not replacements for the finite-pressure bound.

2.2 Intervention channel

This section covers T3 and T4 in `research/core-math.md`.

Before an intervention calculation can be used, the contract must supply a fixed-type action model: actions a , private cost $c(a)$, proxy gain $w \cdot a$, score deficit d , and stakes V . The calculation measures the affordability of proxy movement under that model.

T3. Quadratic Stackelberg wedge. In the one-dimensional noiseless threshold toy with quality Q , action $a \geq 0$, private cost $\frac{a^2}{2\kappa}$, threshold t , and stakes V , gaming by a below-threshold unit is privately worthwhile exactly when $t - Q \leq \sqrt{2\kappa V}$.

The wedge $\sqrt{2\kappa V}$ is a signature of this quadratic toy. It is not a universal intervention law and should not be read as an RLHF, benchmark, or organizational theorem unless the action, cost, stakes, and pass condition have been declared.

T4. Convex score-deficit budget. With finite-dimensional action space, closed proper convex cost c , linear proxy gain $w \cdot a$, and regularity for convex duality, define $m(d) = \inf_a \{c(a) : w \cdot a \geq d\}$. Then

$$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$$

. Gaming under stakes V is feasible exactly when $m(d) \leq V$ in this declared private-cost model.

Cost minimization and hidden-welfare assessment answer different questions. For example, if two proxy channels have equal private cost and equal score weight but hidden harm vector $(M, 0)$, cost minimization splits effort while hidden harm grows with M . The budget licenses private affordability; welfare requires a hidden harm functional.

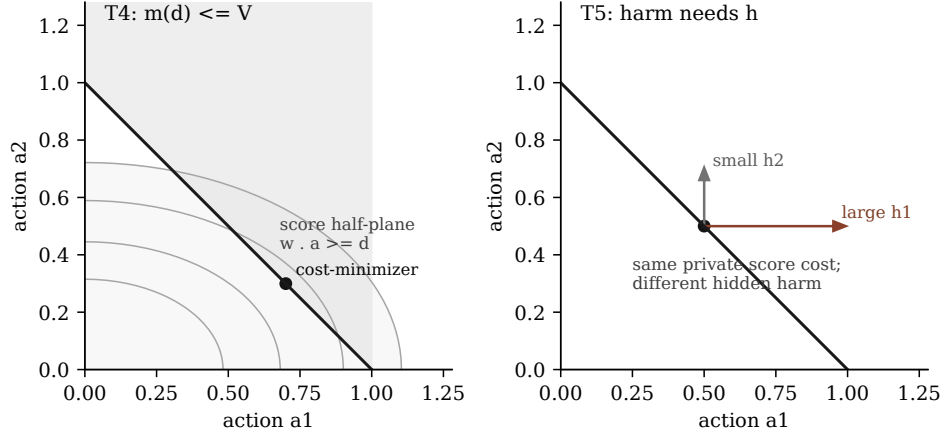


Figure 3: T4 locates the private cost-minimizing action for a score deficit. T5 then requires hidden harm exchange rates before that action can be interpreted as welfare movement.

2.3 Multidimensional scorecards — the keeper

This section covers T5 in `research/core-math.md`.

The scorecard calculation needs more than “more metrics.” It needs an additive score, measured channel set M , weights w_j , separable quadratic costs $\frac{a_j^2}{2\kappa_j}$, fixed score deficit d , and linear hidden harm $\sum_{j \in M} h_j a_j$.

T5. Additive exchange-rate iff. The fixed-deficit per-agent hidden harm of the cost-minimizing action is

$$H_{M(d)} = d \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$

Fixed-deficit harm is conserved across compared active measured sets exactly when $h_j = c w_j$ on those active channels.

This is the most exportable scorecard result. It says what must be checked before saying whether adding or removing a measured channel helped, hurt, or only re-routed harm: the hidden harm per score unit has to be declared or estimated. Holmstrom–Milgrom multitasking is the closest economics precedent, but the contract here makes the hidden-harm exchange rate explicit.

The result is fixed-deficit and per-agent. Population entry is separate: lowering the private cost of reaching the score can recruit more units into gaming even when each fixed-deficit gamer has conserved harm. Conjunctive aggregation is separate too; requiring every measured component to clear a bar can make harm grow with the number of components.

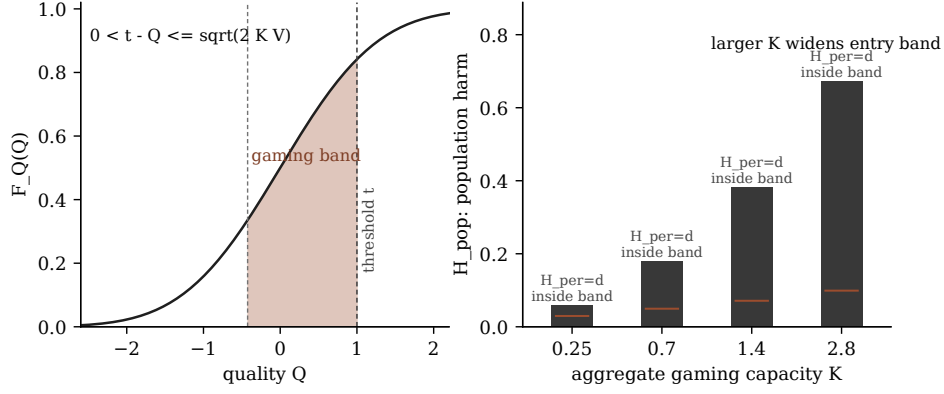


Figure 4: The gaming band separates fixed-deficit per-gamer harm H_{per} from population harm H_{pop} . More capacity widens entry without by itself changing the per-gamer exchange-rate formula.

2.4 Adaptive hardening — narrow but real

This section covers T6 in `research/core-math.md`.

The hardening result is narrow by design. The contract is fixed finite measured set M , fixed deficit d , fixed stakes V , fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone hardening of capacities $\kappa_{j,t}$.

T6. Deterministic adaptive-hardening capacity boundary. Let

$$S_{t(M)} = \sum_{j \in M} \kappa_{j,t} w_j^2$$

and $T = \frac{d^2}{2V}$. At time t , fixed-deficit gaming is feasible exactly when $S_{t(M)} \geq T$. Hardening reaches no-gaming exactly when $S_{t(M)} < T$. A progress-aware largest-action multiplicative rule terminates in finite time when channels are finite, positive-weight, and floor capacity satisfies $S_{\text{floor}(M)} < T$.

Nothing broader is licensed. The theorem does not cover stochastic observation, arbitrary hardening rules, changing measured sets, changing deficits, changing stakes, nonconvex costs, shared bottlenecks, cycles, welfare comparison, or policy optimality. The immediate anti-transfer is noisy observation: once the update rule can chase noise, the deterministic threshold is no longer a stopping theorem.

2.5 Response shape, conditionally

The project did not prove a generic residual-shape law. What survives is a conditional taxonomy:

- Quadratic costs select the minimum-cost direction, often proportional to Cw in a smooth unconstrained model.
- Fixed charges create entry thresholds and active-set comparisons.
- Caps produce spillover only after a channel saturates.
- Low-rank affordances constrain response to an image, but do not choose a basis-invariant “simple” direction by themselves.
- Search-prior claims require a coding language or search process fixed before the failure is observed.

These are not theorem transfers between domains. They are ways to turn a response-shape conjecture into a declared model with falsifiers.

3 Across Disciplines: Primitive Attribution

3.1 Genealogy

Goodhart, Campbell, Strathern, and Manheim–Garra-brant are genealogy, not proof sources for the calculations above. Goodhart’s original macroeconomic warning concerns policy-contaminated regularities [7]. Campbell’s warning concerns social indicators under decision pressure [2]. Strathern’s formulation supplies the familiar compressed warning [20]. Manheim and Garra-brant provide a useful cause taxonomy [13].

This book does not replace that genealogy with a grander taxonomy. It uses the genealogy to ask which primitives are present: selection, response kernel, action cost, proxy/target separation, aggregation, hidden value, and evidence standard.

3.2 Formal analogues, primitive by primitive

The closest formal analogues supply some primitives and omit others. El-Mhamdi–Hoang and Majka–El-Mhamdi are scalar tail-conditioned selection anchors [5, 12]. Hardt et al. strategic classification supplies costly feature-change intervention [8]. Perdomo et al. performative prediction supplies deployment-induced response kernels [15]. Skalse et al. supply proxy/target and RL-specific response-kernel tools [11, 18]. Holmstrom–Milgrom supplies the closest multitask incentive analogue for aggregation and action costs [10]. Cawley–Talbot, reusable holdout, leaderboard work, and optimizer’s curse supply selection and evidence standards [1, 3, 4, 16, 17, 19]. Pan and Gao provide empirical response-geometry candidates, not plug-in parameters [6, 14].

3.3 What each discipline contributes vs. omits

Source family	Primitives supplied	Primitives omitted	Licensed transfer
Tail-conditioned Goodhart	Selection, scalar proxy/target, tail assumptions.	Vector aggregation, intervention, hidden welfare.	Sharp scalar asymptotics where the tail contract holds; otherwise only a reweighting analogy.
Strategic classification and performative prediction	Response kernels, costly action or endogenous deployment.	General hidden value, scorecard exchange rates, marginal identification.	Intervention framing after action/cost or deployment response is declared.
RL reward hacking	Proxy/target separation, MDP oc-	Non-MDP domains, empirical	RL-specific response calculations

Source family	Primitives supplied	Primitives omitted	Licensed transfer
	occupancy geometry, optimization pressure.	welfare, generic cost parameter.	only inside the MDP/occupancy contract.
Multitask incentives	Aggregation and multidimensional effort/cost.	The book’s value-metric contract, non-LEN settings unless separately shown.	External precedent for the score-card exchange-rate question, not a proof of this book’s theorem.
Adaptive holdout and leaderboard hygiene	Selection/evidence standards for repeated benchmark use.	Fixed-type fine-tuning, contamination, welfare, action cost.	Evidence discipline for selection over fixed candidates and public/private score interpretation.
Reward overoptimization empirics	Candidate search and optimization-pressure variables.	A defended mapping from model size, KL, data access, or prize to κ or V .	Hypothesis generators for response geometry; no automatic theorem transfer.

3.4 Reduction, not unification

No single equation unifies these literatures. The framework’s claim is narrower: each formalism fills different fields of the response-modeling contract. The contract says which primitive is being borrowed and which transfer fails. A source about adaptive holdouts does not identify hidden welfare. A source about costly feature change does not identify value weights. A source about multitask contracts does not make “more metrics” good or bad in a new application.

4 Cases as Stress Tests

4.1 MMLU

MMLU is useful because it is not a single mechanism [9]. A public benchmark score can induce fixed-checkpoint model selection, repeated adaptive testing, prompt search, finetuning on benchmark-like data, contamination, reward-model overoptimization, or genuine capability transfer.

The contract separates these mechanisms. Fixed-checkpoint comparison is a selection problem: candidate checkpoints are types, the score is a proxy, and the hidden target is performance away from the benchmark. Finetuning, prompt search, contamination, or reward optimization is intervention/search only after an action/search geometry is declared. MMLU alone supplies neither κ nor stakes V , and it does not supply hidden welfare. T3/T4 apply only after a cost model and score deficit are declared. T5 applies only after benchmark components are modeled as additive channels with harm exchange rates.

The stress test is whether the evidence can distinguish fixed-candidate selection from post-exposure adaptation. Necessary observations include benchmark access, training data lineage, prompt/search budget, repeated-query history, transfer to non-MMLU tasks, and failure modes that move when the benchmark format changes.

4.2 Hospital readmissions

For a readmission scorecard, score improvement is compatible with hospitals leaving the comparison pool, coding changes, observation-status changes, delayed admissions, patient avoidance, better follow-up care, or mixtures.

The auditable primitives are concrete: effective score weights w , stakes V , response ease κ for coding, discharge timing, follow-up, and patient selection channels, hidden harm rates h , and signal adequacy γ for the measured components. If these are unavailable, the design consequence is not “assume the score is bad.” It is: do not use the toy diagnostic, collect action traces, monitor hidden patient outcomes where possible, pilot or lower leverage, and mark the missing primitive.

This is an evidence contract, not policy advice. The framework can say that aggregate readmission movement is insufficient to credit patient-welfare improvement. It cannot rank hospital policies without the clinical and value model the contract explicitly requires.

4.3 Scientific metrics

Publication counts, citations, grants, venue prestige, and rankings are already covered by responsible-metrics warnings such as DORA and the Leiden Manifesto. The framework does not claim novelty for the warning that crude metrics can distort science.

The stress test is discriminator value. A scientific scorecard application must separate selection over researchers, labs, fields, and institutions from fixed-researcher response. It must distinguish harmful proxy manufacture from harmless metadata/

discoverability repair and from genuine improvements in methods, robustness, replication value, useful synthesis, or long-run uptake.

The contract refuses broad research-value inference from score movement. If an institution cannot declare the target, aggregation, action traces, and hidden value model before interpreting a score rise, the framework returns “no licensed hidden-value claim.”

5 Practical Implications

5.1 What to do before crediting score movement

Wentworth’s practical warning is that experiments often measure something other than what the experimenter thinks they are measuring: an unexpected confounder, a secondary channel, a sampling artifact, or an operational detail that happens to track the named quantity [21]. The proposed repair is not to stare harder at the headline metric. It is to measure many auxiliary traces so the unexpected channel has somewhere to show up.

This book agrees with the first half and adds a second step. Auxiliary traces are necessary for discovery, but they are not sufficient for interpretation. They must feed a declared response contract. The combined rule is: collect the firehose for discovery, then use the contract for claim licensing. Without the firehose, the analyst misses the confounder. Without the contract, the analyst has many traces and still no disciplined statement of which score movement counts as selection, fixed-type response, proxy repair, harmful gaming, real improvement, or a mixture.

The practical workflow is therefore not to ask first, “did the score improve?” Ask instead: what changed, who changed, by what channel, at what cost, with what hidden target evidence, and what observation would make this interpretation fail? A score rise is an observation to be explained, not a conclusion to be credited.

Audit field	Entry before crediting score movement
Observed score movement	Magnitude, timing, affected units, score components, and whether the movement is marginal, thresholded, ranked, or aggregate.
Candidate mechanisms	Selection, fixed-type response, proxy repair, harmful gaming, real improvement, or an explicit mixture.
Required discriminator traces	Repeated-type evidence, exposure timing, action logs, search or query budget, component-level score changes, participation changes, and off-score outcomes that separate nearby mechanisms.
Hidden value/harm evidence	Declared target coordinates, value weights, harm rates, residual measures, or domain outcomes; if absent, mark the hidden-value claim unlicensed.
Missing primitives and operational consequence	The undeclared type space, baseline law, response channel, action cost, aggregation rule, or hidden value model, plus the resulting consequence: collect traces, lower leverage, pilot, or withhold the claim.
Licensed claim	The narrow statement supported by the declared contract: for example, pure-selection drift envelope, private affordability under a cost model, exchange-rate diagnostic, or no hidden-value conclusion.

Audit field	Entry before crediting score movement
Blocked claim	The stronger statement not supported: welfare improvement, target improvement, policy optimality, channel identification, or generic anti-metric advice.
Contract-failure condition	A concrete observation that would defeat the interpretation, such as action traces inconsistent with the named channel, hidden outcomes moving opposite the claimed target, or theorem conditions failing.

This sheet is intentionally smaller than the full application template. Its job is to slow down the common interpretive jump. If the observed movement is a readmission decline, a benchmark rise, a citation increase, or a ranking gain, the first audit question is the same: what response channel could have produced this movement? The second is which traces would discriminate that channel from its nearest rivals. The third is which hidden value or harm evidence is present, and which claim remains blocked if it is absent.

The conclusion can be positive, but it has to be narrow. “The score improved and auxiliary outcomes moved in the declared target direction under stable exposure” is a different claim from “the policy improved welfare.” “The action logs fit a low-cost proxy-repair story” is a different claim from “the metric is safe.” “The selection envelope bounds hidden coordinate drift under the stated law” is a different claim from “selection is harmless.” Practical use of the framework is mostly this discipline of replacing a large conclusion with the smaller one the evidence actually licenses.

6 What the Framework Refuses to Do

6.1 Anti-applications

The framework is least useful when its primitives cannot be stably declared. Concrete anti-applications include:

- A rapidly shifting online platform where type space, measured set, policy exposure, and available actions all change between measurements.
- A black-box institutional ranking where effective weights and thresholds are unobservable and participants cannot tell which action channel is being rewarded.
- A high-stakes public decision where stakeholders demand a welfare verdict but no defensible hidden value model exists.

In these cases, the contract returns “no verdict.” That is not a hedge; it is the framework refusing to convert score movement into causal mechanism or welfare language without the primitives that would make the claim testable.

6.2 Falsifiers

The contract itself is falsifiable as a modeling discipline. A concrete falsifier would be a domain where the primitives are declared in advance, response channel and action traces are observable enough to distinguish the nearby mechanisms, and the licensed calculation systematically predicts the wrong response shape or direction while a simpler score-only rule predicts it correctly.

Examples include:

- In a pure-selection setting with measured L and hidden coordinates, hidden drift repeatedly exceeds the δs_i envelope under the stated moment and absolute-continuity conditions.
- In a fixed finite deterministic hardening setting, with stable M , d , V , weights, separable quadratic costs, and deterministic observation, gaming remains feasible after $S_{t(M)} < \frac{d^2}{2V}$ or stops while $S_{t(M)} \geq \frac{d^2}{2V}$.
- In an additive scorecard with defended w_j , κ_j , and h_j , fixed-deficit per-agent harm is conserved across active measured sets even when the exchange-rate condition $h_j = cw_j$ fails, or fails to be conserved when the condition holds.

If those failures survived audit, the framework would not merely need prose repair; the licensed theorem import would be wrong or the contract fields would not track the response mechanism they were supposed to track.

7 Open Agenda

7.1 The residual-shape conjecture

The signature open problem is Q18: when does repeated proxy repair drive hidden failure toward a predictable residual shape, such as a low-complexity attractor? The current answer is negative unless a mechanism is named first.

Resolving Q18 requires at least five declarations: a response mechanism, a complexity or shape functional fixed before inspection, a policy-update rule, a composition rule for repeated repair, and a failure condition. It would count as progress to prove a sparse, low-rank, or low-description-length attractor inside one of those contracts. It would not count to observe a simple-looking failure after the fact and relabel it as the attractor.

7.2 Composition, identification, and information

Three structural gaps remain.

First, channels compose. Real cases mix selection, fixed-type response, proxy repair, and real improvement. The current contract names mixtures but does not give a general calculus for them.

Second, primitives are not automatically identifiable. Marginal score movement does not identify W_θ versus K_θ , hidden harm, action costs, or aggregation. The framework needs identification toys and evidence thresholds for when a primitive can be treated as declared rather than guessed.

Third, the L^2 selection bound may not be the portable final form. An information-theoretic restatement could travel further if it preserves the distinction between coordinate-explicit drift, declared value metrics, and finite-pressure path behavior.

7.3 Toolkit gap

A practitioner should not have to re-derive the contract every time. The missing toolkit has three parts: a primitive-elicitation protocol, worked exchange-rate audits in real scorecards, and small identification examples that show what observations distinguish selection, intervention, proxy repair, and real improvement.

Until those tools exist, the framework is best read as a claim-license discipline and a theorem inventory. It tells a reader what would have to be true before a Goodhart calculation travels.

References

- [1] Avrim Blum and Moritz Hardt. 2015. The Ladder: A Reliable Leaderboard for Machine Learning Competitions. In *Proceedings of the 32nd International Conference on Machine Learning (Proceedings of Machine Learning Research)*, 2015. PMLR, 1006–1014. Retrieved from <https://proceedings.mlr.press/v37/blum15.html>
- [2] Donald T. Campbell. 1979. Assessing the impact of planned social change. *Evaluation and Program Planning* 2, 1 (1979), 67–90.
- [3] Gavin C. Cawley and Nicola L. C. Talbot. 2010. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *Journal of Machine Learning Research* 11, (2010), 2079–2107. Retrieved from <https://www.jmlr.org/papers/v11/cawley10a.html>
- [4] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. 2015. Generalization in Adaptive Data Analysis and Holdout Reuse. In *Advances in Neural Information Processing Systems*, 2015. Retrieved from <https://arxiv.org/abs/1506.02629>
- [5] El-Mahdi El-Mhamdi and Lê-Nguyen Hoang. 2024. On Goodhart's Law, with an Application to Value Alignment. Retrieved from <https://arxiv.org/abs/2410.09638>
- [6] Leo Gao, John Schulman, and Jacob Hilton. 2023. Scaling Laws for Reward Model Overoptimization. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, 2023. PMLR, 10835–10866. Retrieved from <https://proceedings.mlr.press/v202/gao23h.html>
- [7] Charles A. E. Goodhart. 1975. Problems of Monetary Management: The U.K. Experience. *Papers in Monetary Economics, Volume I*.
- [8] Moritz Hardt, Nimrod Megiddo, Christos Papadimitriou, and Mary Wootters. 2016. Strategic Classification. In *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS)*, 2016. 111–122.
- [9] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring Massive Multitask Language Understanding. In *International Conference on Learning Representations*, 2021. Retrieved from <https://arxiv.org/abs/2009.03300>
- [10] Bengt Holmström and Paul Milgrom. 1991. Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design. *The Journal of Law, Economics, and Organization* 7, Special Issue (1991), 24–52. https://doi.org/10.1093/jleo/7.special_issue.24
- [11] Jacek Karwowski, Oliver Hayman, Xingjian Bai, Klaus Kiendlhofer, Charlie Griffin, and Joar Skalse. 2023. Goodhart's Law in Reinforcement Learning. Retrieved from <https://arxiv.org/abs/2310.09144>

- [12] Adrien Majka and El-Mahdi El-Mhamdi. 2025. The Strong, Weak and Benign Goodhart's Law. An Independence-Free and Paradigm-Agnostic Formalisation. Retrieved from <https://arxiv.org/abs/2505.23445>
- [13] David Manheim and Scott Garrabrant. 2018. Categorizing Variants of Goodhart's Law. *arXiv preprint arXiv:1803.04585* (2018).
- [14] Alexander Pan, Kush Bhatia, and Jacob Steinhardt. 2022. The Effects of Reward Misspecification: Mapping and Mitigating Misaligned Models. In *International Conference on Learning Representations*, 2022. Retrieved from <https://openreview.net/forum?id=JYtwGwIL7ye>
- [15] Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünnér, and Moritz Hardt. 2020. Performative Prediction. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. 7599–7609.
- [16] Benjamin Recht, Rebecca Roelofs, Ludwig Schmidt, and Vaishaal Shankar. 2019. Do ImageNet Classifiers Generalize to ImageNet?. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, 2019. PMLR, 5389–5400. Retrieved from <https://proceedings.mlr.press/v97/recht19a.html>
- [17] Rebecca Roelofs, Vaishaal Shankar, Benjamin Recht, Sara Fridovich-Keil, Moritz Hardt, John Miller, and Ludwig Schmidt. 2019. A Meta-Analysis of Overfitting in Machine Learning. In *Advances in Neural Information Processing Systems*, 2019. Retrieved from <https://arxiv.org/abs/1906.02168>
- [18] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krashenninikov, and David Krueger. 2022. Defining and Characterizing Reward Gaming. In *Advances in Neural Information Processing Systems*, 2022.
- [19] James E. Smith and Robert L. Winkler. 2006. The Optimizer's Curse: Skepticism and Postdecision Surprise in Decision Analysis. *Management Science* 52, 3 (2006), 311–322. <https://doi.org/10.1287/mnsc.1050.0451>
- [20] Marilyn Strathern. 1997. 'Improving ratings': audit in the British University system. *European Review* 5, 3 (1997), 305–321.
- [21] John Wentworth. 2022. You Are Not Measuring What You Think You Are Measuring. Retrieved from <https://www.lesswrong.com/posts/9kNxxhKWvixtKW5anS/you-are-not-measuring-what-you-think-you-are-measuring>